

Fundamental Noise and Gravitational-Wave Sensitivity of the Laser Interferometer Lunar Antenna (LILA)

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Abstract. The Earth’s Moon presents a uniquely advantageous environment for detecting astrophysical gravitational waves (GWs) in the frequency range of millihertz to decihertz. Unlike Terrestrial GW detectors, the quiet seismic environment of the Moon does not impede detection in this band; in fact the ground motions of the Moon will be excited by GWs, making the Moon a resonant amplifier at low frequencies. The Laser Interferometer Lunar Antenna (LILA) mission aims to be limited by thermal Brownian noise in its optics across most target frequencies. By taking advantage of the lunar normal mode resonances, we show that the initial phase of the mission, iLILA, achieves the GW sensitivity required to study astrophysical sources through the millihertz to decihertz range. The advanced phase of the mission, LILA Observatory Lunar Deformation Channel, would increase GW sensitivity to the cosmological horizon in this band.

1. Introduction

In recent years there has been a surge in interest in Lunar science, in parallel with increasing capabilities of private space enterprises and national interest in crewed missions to the Moon. One of the more intriguing possibilities for Lunar science is in gravitational-wave (GW) astronomy. Due to its quiet seismic environment and natural hard vacuum, the Moon is potentially a far more productive environment than Earth for operating GW detectors. Possible detection schemes range from pure seismic accelerometers that treat the Moon as a mechanical resonant GW antenna [1], to long-baseline interferometers with optical elements fixed on the Moon’s surface [2], to fully-suspended interferometric antennas [3] akin to Terrestrial observatories such as LIGO and Virgo.

The Lunar Interferometer Laser Antenna (LILA) [4] proposes a phased approach to building a laser-based observatory for gravitational-wave detection and Lunar science. The first proposed mission is the LILA Autonomous Interferometry Technology Demonstration AITD, which is not intended to perform actual GW

observations. The initial science mission iLILA would have an L-shaped unsuspended laser interferometer with 1km arms, while a follow-on LILA Observatory would expand to a 40km triangular observatory including a Lunar Deformation Channel LDC with unsuspended optics, and a Geodesic Displacement Channel GDC with suspended optical elements. This paper presents noise and sensitivity estimates for the unsuspended LDC component of the iLILA and LILA Observatory.

The iLILA mission specifically will consist of a primary corner module and secondary end modules, deployed from a single lander and Lunar Terrain Vehicle. Details about site selection can be found in [5]. The corner module contains the laser emitter, receiver, and associated electronics and optics (laser frequency stabilization, input and output mode-cleaners). The end modules are much simpler, containing only a reflector (a steerable mirror or coherent retroreflector) to maximize the return signal. All modules will have sunshields and insulated enclosures to regulate temperature, and short beamtubes with baffles and electrostatic precipitators to reduce stray light and dust.

The three stations will be operated either as a Michelson interferometer or as a pair of single-arm strainmeters, depending on the achievable laser stability (see Section 2.4). In either case, the equal-arm L shape has the benefit of maximizing the effective baseline (total optical path) within the constraints of a maximum sortie range of the Lunar Terrain Vehicle that will be used to deploy the end stations. An interferometric detector has the advantage that it cancels most of the laser phase noise in the antisymmetric port, while a pair of strainmeters would provide two independent baselines for measuring Lunar deformations other than those induced by gravitational waves.

As noted, the base design will contain no suspension system; all elements will be fixed to a monolithic optical bench anchored to the Lunar surface. It will not be cryogenic, but will be maintained at a constant temperature similar to the ambient average temperature ~ 300 K.

2. Noise Sources

Dominant fundamental noise sources for Terrestrial GW detectors are typically some combination of seismic noise (ground motion), suspension thermal noise, thermal noise in optics and coatings, and quantum readout noise in the laser system. Other noise sources include laser phase noise, scattered light, Newtonian (local gravitational) noise, and technical noise. We will consider these in turn.

2.1. Seismic Noise

The Earth is a seismically noisy environment, due to geological processes, along with ground motions produced by waves, wind, and atmospheric pressure loading. Anthropogenic noise can also be observed near cities. In particular, there is significant continuous noise at all points on the Earth from 0.05 to 1 Hz caused by the oceans, referred to as microseismic noise [6]. Terrestrial GW detectors isolate their test masses (mirrors) from these motions through elaborate suspension systems. By contrast, the Moon is a seismically quiet environment, with a largely inactive interior and few surface phenomena that would produce ground movement. Apollo seismometers did record many transient “moonquakes” due to tidal stress, meteorite impacts, and local thermal changes, but the continuous seismic background consistently remained below

the noise floor of the instruments, which was just above $10^{-10}\text{m/s}^2/\sqrt{\text{Hz}}$ at its peak sensitivity near 0.5 Hz [7, 8], and has been modeled to be significantly lower if driven by micrometeorite impacts [9].

In the absence of direct measurements of the Lunar seismic background, we will consider here the sensitivity of an unsuspended strainmeter to gravitational waves without such background. Another paper [10] explores the use of LILA strainmeters for Lunar geophysics (*selenophysics*). Furthermore, in the millihertz band, we assume that *gravitational waves* will themselves excite seismic motions in excess of any such background; see Section 3. Thus in each case we treat ground motion as a potential *signal* rather than a noise source.

2.2. Thermal Noise

Lacking suspensions, the proposed strainmeter will similarly lack suspension thermal noise, so we need only consider Brownian thermal noise for optical surfaces along the beam path. A standard analysis [11] using fluctuation-dissipation theorem gives the power spectral density $S_x(f)$ in the position x of an optical surface to be:

$$S_x(f) = \frac{2k_B T(1 - \sigma^2)\phi_{\text{eff}}}{\pi^{3/2}fwY} \quad (1)$$

where k_B is Boltzmann’s constant, T is the temperature, σ and Y are the Poisson ratio and Young’s modulus respectively of the optical substrate material, w is the half-width of the beam spot (assumed Gaussian), and ϕ_{eff} is the effective loss angle of any dissipative mechanical processes. However, this is an idealized case where dissipation is only through the bulk of the substrate material. Thermal noise will be dominated by the most mechanically lossy (dissipative) element contributing to the optical path, which may be the mirror coatings, mountings, optical bench, etc. In this analysis we treat the optics as rigidly attached to the Lunar surface; a more detailed future analysis would need to model the mounting of the optical bench to the lander, attachment of the lander to Lunar regolith or (preferably) bedrock, and Brownian noise from the Moon itself. These effects in practice are usually determined empirically [12] or through finite-element modeling [13]. From [13] we treat the following noise estimate as potentially achievable:

$$S_x(f) = (6 \times 10^{-32}\text{m}^2/\text{Hz}) \left(\frac{T}{300\text{K}} \right) \left(\frac{f}{\text{Hz}} \right)^{-1} \quad (2)$$

Since the basic strainmeter does not employ high-gain optical cavities on its arms, it will be dominated by the narrowest beam size on the lossiest element anywhere on the optical path: this model assumes the beam size is 0.5 mm (a typical laser aperture), and the most lossy elements are the piezoelectric actuators on the mirrors; anchoring to Lunar rock might incur similar loss factors. Ideally it would be preferable to use electro-optic modulators rather than piezoelectrics for beam shaping and phase control, which would be both lower-voltage and mechanically stiffer. Optimistically, assuming that we can reach thermal noise comparable to the optical coatings in [13], then our position noise could be as low as:

$$S_x(f) = (1.2 \times 10^{-33}\text{m}^2/\text{Hz}) \left(\frac{T}{300\text{K}} \right) \left(\frac{f}{\text{Hz}} \right)^{-1} \quad (3)$$

Because the actual prefactor for a Lunar mission will require numerical or experimental modeling of the final optical system, our noise curves will include it as a free parameter $S_{x,T}(1\text{Hz})$ with “canonical” values ranging from 1.2 to $60 \times 10^{-33}\text{m}^2/\text{Hz}$.

2.3. Quantum Noise

Quantum noise refers to noise resulting from the quantization of the electromagnetic field of the laser. At high frequencies this is called *shot noise*, the limitation of phase readout at the photodetector due to stochastic reception of individual photons. This gives fluctuations in the measured optical path length x :

$$S_x(f) = \frac{\hbar c \lambda}{2\pi \eta P} = (5.4 \times 10^{-32} \text{m}^2/\text{Hz}) \left(\frac{\eta P}{100 \text{mW}} \right)^{-1} \quad (4)$$

where $\lambda = 1064 \text{nm}$ is the wavelength of the laser, P is the laser power, and η is the quantum efficiency (fraction of emitted photons detected at the photoreceptor), including losses along the optical path. (See e.g. [14] Eq. (4.4).)

We note that for $\eta P \gtrsim 100 \text{mW}$ (a reasonable readout power), we will not be dominated by shot noise below 1 Hz for the more conservative thermal noise estimates. However, if we pursue the more aggressive thermal noise limits, then we will also want to increase the power on the photoreceptor: $\sim 500 \text{mW}$ would be a practical upper limit. We also note that rigidly-mounted optics will not be subject to the low-frequency *radiation pressure noise* that limits the performance of high-power laser interferometers with suspended optics.

2.4. Other Noise

We list here a few noise sources that we have considered and believe can be neglected or mitigated in order to estimate the detector's ultimate sensitivity.

Stray light from objects other than the distant retroreflector will introduce noise in range measurements. However, a well-designed output mode-cleaner should be able to reject off-axis light, leaving us susceptible only to dust scattering within the beam path. The LADEE mission [15] measured dust densities of $N \sim 3 \times 10^{-3} \text{m}^{-3}$ below 40 km with particle sizes $a \lesssim 0.7 \mu\text{m}$. The relatively uniform density profile below 40 km suggests typical ballistic grain speeds $v \sim 300 \text{m/s}$. Using scattering formulae (12), (13), and (14) from [16] (and adjusting to our notation), the estimated scattering noise is:

$$\begin{aligned} S_x(f) &\approx \frac{16\pi^2(n^2 - 1)^2}{9} \frac{LN a^6}{wv} \\ &\approx (1.6 \times 10^{-35} \text{m}^2/\text{Hz}) \left(\frac{L}{5 \text{km}} \right) \left(\frac{N}{3 \times 10^{-3} \text{m}^{-3}} \right) \\ &\quad \times \left(\frac{a}{0.7 \mu\text{m}} \right)^6 \left(\frac{v}{300 \text{m/s}} \right)^{-1} \left(\frac{w}{1 \text{cm}} \right)^{-1} \end{aligned} \quad (5)$$

where $n \approx 1.5$ is the grain index of refraction and w is the beam width along the arm. This estimate places scattering noise well below shot noise.

Laser phase noise encourages the use of equal-arm interferometers, since it cancels when light from the arms combine. In principle a sufficiently frequency-stabilized laser would allow the two arms to be operated as independent strainmeters. This would require frequency noise of order $d\nu/\nu \sim dL/L \sim 10^{-24}/\sqrt{\text{Hz}}$, or $d\nu \sim 3 \times 10^{-10} \text{Hz}/\sqrt{\text{Hz}}$, which would entail an on-board high-finesse stabilization cavity. More conservatively, an unstabilized laser with frequency noise ~ 4 orders of magnitude higher would require placement of the arm retroreflectors within a few centimetres of arm length equality, which should be achievable. The final iLILA design

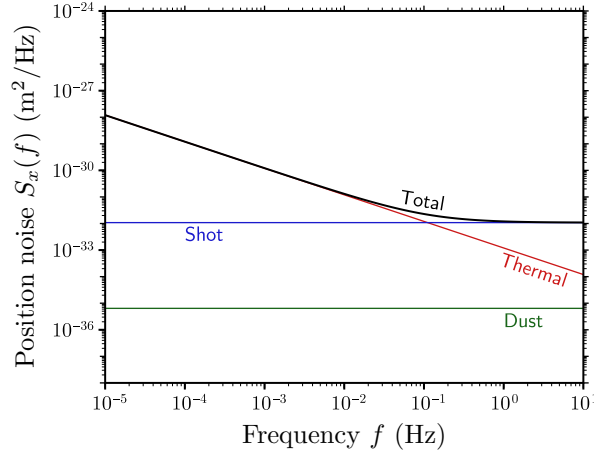


Figure 1. Position noise contributions from thermal noise, shot noise, and dust scattering for iLILA (1 km L-shaped) detector.

will be chosen to ensure that laser noise will not dominate the GW sensitivity with the available flight hardware.

Newtonian gravitational noise from nearby moving masses is a concern for 3rd Generation Terrestrial detectors at low frequencies. However, this is because the gravitational coupling bypasses the suspension system that otherwise isolates the test masses from ground motion. For an unsuspended system the Newtonian gravitational perturbations will always be less than the ground motions, which we are neglecting as a noise source.

“Technical noise” refers to various measurement errors and perturbations introduced by feedback and control loops, electronic and computational artifacts, etc. For purposes of this analysis we will treat these as solvable problems rather than inimitable noise.

3. Response Function

Following [17], we identify two mechanisms by which a gravitational wave may excite a Lunar-based detector: an *inertial regime*, in which the Moon acts like a fluid body, and a *mechanical regime*, in which the Moon behaves like an elastic oscillator. These two regimes are distinguished by the mechanical response of the Moon QL_{eff} , where L_{eff} is the effective lengthscale that interacts with the gravitational wave (of order the wavelength of a seismic wave of the same frequency), and Q is the Moon’s mechanical quality factor at that frequency. When $QL_{\text{eff}} \ll R_{\text{Moon}}$, the Moon behaves inertially; when $QL_{\text{eff}} \gg R_{\text{Moon}}$, the Moon behaves resonantly.

3.1. Lunar Inertial Response

The inertial regime is perhaps the easiest to understand. In this limit, the surface responds to the gravitational wave like a collection of free masses, with no proper acceleration in the horizontal direction. However, the change in distance between a

pair of masses separated by a distance L goes as:

$$\delta x = \frac{1}{2}hL \quad (6)$$

where h is the projection of the gravitational wave on the arm axis. For three masses at the corner and ends of two arms with an opening angle $\theta > 30^\circ$, we can improve on this by measuring the changing *difference* in arm length, giving:

$$\delta x = hL \sin \theta \quad (7)$$

This is the sensitivity to an *optimally oriented* gravitational wave. For a single-arm strainmeter, this is a $+$ polarized wave incident at right angles to the axis L . For a two-arm configuration, it is a wave incident from the zenith, whose $+$ polarization axes are symmetric about (i.e. have the same bisector as) the two arms. We note that the “beam pattern” response for non-optimally-incident waves is conventionally included in the $h(t)$ of the source object or population rather than the instrument sensitivity curve: the reduction in response is of order 0.4, and will differ by tens of percent depending on the beam pattern and assumptions about the source distribution.

The resulting strain sensitivity is simply:

$$S_h(f) = \frac{S_x(f)}{L^2 \sin^2 \theta} \quad (8)$$

where $\theta \approx 90^\circ$ for iLILA and 60° for LILA Observatory. If operating as a single-arm strainmeter we replace $\sin \theta$ with $\frac{1}{2}$.

3.2. Lunar Mechanical Response

When $QL_{\text{eff}} \gg R_{\text{Moon}}$, the mechanical oscillations excited in the Moon produce proper accelerations whose displacements begin to exceed the relative displacements of free masses. For frequencies near the Lunar normal modes, this can lead to significantly higher responses.

There has been extensive modeling of this response in the context of seismometer-based detectors, since such detectors *only* respond to proper accelerations. We take as our starting point the horizontal response function in Figure 5 of [18]. The precise shape of this response curve is dependent on numerical simulations, based on a plausible but hypothetical Lunar model, so we first consider the broad behaviour rather than the specific detailed structure.

Generally, their base response (which we can identify with L_{eff}) goes as a power law from 10^3 m per unit strain at 10^0 Hz, to 10^6 m per unit strain at 10^{-3} Hz. However, the response has peaks at mode frequencies where the Moon acts as a resonant amplifier, increasing the response by a factor Q of up to 10^2 at 1 Hz, to 10^4 at 1 mHz.

$$L_{\text{eff}} \approx 10^3 \text{ m} \left(\frac{f}{\text{Hz}} \right)^{-1}, \quad Q \sim 10^2 \left(\frac{f}{\text{Hz}} \right)^{-2/3}. \quad (9)$$

Realistically, partial melting in the deep mantle of the Moon will limit the low-frequency Q to $\lesssim 500$ or so, and scattering due to the “megaregolith” (the heavily-fractured upper crust and mantle to a depth of ~ 20 km [19]) will reduce Q sharply above ~ 0.1 Hz, due to the loss of mode coherence for seismic wavelength less than of order the megaregolith depth.

More precisely, the Lunar displacement response $r(f)$ is the result of *all* the excited normal modes, each having the form of a mechanical oscillator with characteristic frequency f_m and quality Q_m :

$$r(f) = \left| \sum_m \frac{f^2 l(f)}{f_m^2 - f^2 + i f_m f / Q_m} \right|, \quad l(f) \approx 30 \text{ m} \left(\frac{f}{\text{Hz}} \right)^{-2} \quad (10)$$

Note that $l(f)$ here is *not* the same as L_{eff} : the effective interaction lengthscale at any frequency includes the upper tail response of all the normal modes below that frequency (which number increases linearly with f). The expression in Equation (10) is an approximate fit to the numerical result in Figure 5 of [18].

For our analysis we will use the formula above to approximate the Lunar mechanical response, but will take our spectrum of normal modes from [20] and calculated with the code [21]. This gives approximately 870 quadrupole normal modes between the fundamental (1 mHz) and 0.7 Hz. We further apply a smooth cutoff on Q_m for $f_m \gtrsim 0.05$ Hz to account for megaregolith scattering. See Section 4 for more details.

These modes will all have $l = 2$ quadrupolar *angular* shape (albeit with different *radial* eigenfunctions depending on frequency). Since a strainmeter of length $L \ll R_{\text{Moon}}$ only measures a small fraction of an angular wavelength, the *differential* horizontal displacement will be reduced by a factor kL where $k = (l + \frac{1}{2})/R_{\text{Moon}}$ is the angular wavenumber. As with the inertial response, a gravitational wave that induces a shear in orthogonal directions will produce a differential displacement up to twice this amount:

$$\delta x = \frac{5r(f)L \sin \theta}{R_{\text{Moon}}} h \quad (11)$$

We note that at high frequencies, as $r(f)$ (i.e. QL_{eff}) decreases, this displacement will eventually be less than the inertial displacement $\delta x = Lh \sin \theta$; this corresponds to the transition to the inertial regime, above. Thus:

$$S_h(f) \approx \frac{S_x(f)}{L^2 \sin^2 \theta} \left[1 + \frac{25r(f)^2}{R_{\text{Moon}}^2} \right]^{-1} \quad (12)$$

3.3. Low-Frequency Cutoff

At very low frequencies, below the Moon's fundamental mode $f_0 \approx 1$ mHz, there should be no inertial regime: we expect the Moon to behave as a solid elastic object rather than a fluid, with displacement proportional to the driving force $\delta x \propto L \nabla g$ where $\nabla g = \frac{1}{2} \ddot{h} = \frac{1}{2} \omega^2 h$ is the tidal gravity gradient. We model this by removing the fundamental mode response $m = 0$ from the sum in Equation (10) and apply it as a multiplicative factor to the overall response. Since each mechanical resonance has the effect of a high-pass filter, this leaves the higher-frequency mode response unchanged, while giving the required $\propto f^{-2}$ behaviour for $f \ll f_0$.

$$S_h(f) \approx \frac{S_x(f)}{L^2 \sin^2 \theta} \left| \frac{f_0^2 - f^2 + i f_0 f / Q_0}{f^2} \right|^2 \times \left[1 + \frac{25l(f)^2}{R_{\text{Moon}}^2} \left| \sum_{m \neq 0} \frac{f^2}{f_m^2 - f^2 + i f_m f / Q_m} \right|^2 \right]^{-1} \quad (13)$$

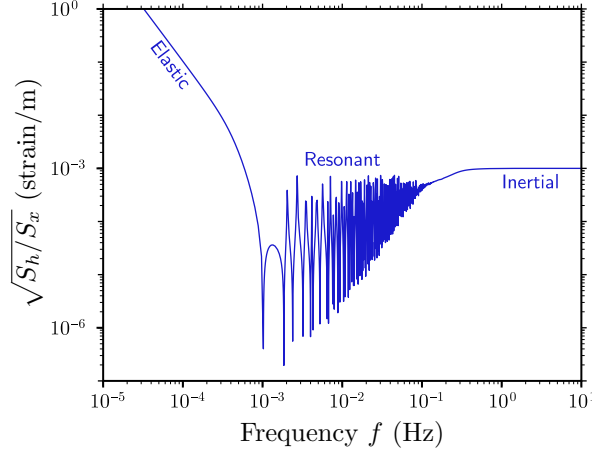


Figure 2. Transfer function (inverse of Lunar response) to gravitational-wave excitation, indicating *inertial* (fluid), *resonant* (excited normal modes), and *elastic* (solid deformation) regimes, for iLILA (1 km L-shaped) detector.

4. Gravitational-Wave Sensitivity

Combining Equations (3), (4), and (13), taking $R_{\text{Moon}} = 1740 \text{ km}$, $l(f) = \frac{3}{0} \text{ m}(f/\text{Hz})^{-2}$, we get the following estimated LILA noise power spectral density $S_h(f)$ and characteristic sensitivity $h_c = \sqrt{f S_h(f)}$:

$$\begin{aligned}
 S_h &= (4 \times 10^{-41} / \text{Hz}) \left(\frac{L \sin \theta}{5 \text{ km}} \right)^{-2} \\
 &\times \left[\left(\frac{S_{x,T}(1 \text{ Hz})}{10^{-33} \text{ m}^2 / \text{Hz}} \right) \left(\frac{T}{300 \text{ K}} \right) \left(\frac{f}{\text{Hz}} \right)^{-1} + 54 \left(\frac{\eta P}{100 \text{ mW}} \right)^{-1} \right] \\
 &\times \left| \frac{1}{T_0(f)} \right|^2 \left[1 + (7.4 \times 10^{-9}) \left(\frac{f}{\text{Hz}} \right)^{-4} \left| \sum_{m \neq 0} T_m(f) \right|^2 \right]^{-1} \quad (14)
 \end{aligned}$$

$$\begin{aligned}
 h_c &= (6.3 \times 10^{-21}) \left(\frac{L \sin \theta}{5 \text{ km}} \right)^{-1} \\
 &\times \left[\left(\frac{S_{x,T}(1 \text{ Hz})}{10^{-33} \text{ m}^2 / \text{Hz}} \right) \left(\frac{T}{300 \text{ K}} \right) + 54 \left(\frac{\eta P}{100 \text{ mW}} \right)^{-1} \left(\frac{f}{\text{Hz}} \right) \right]^{1/2} \\
 &\times \left| \frac{1}{T_0(f)} \right| \left[1 + (7.4 \times 10^{-9}) \left(\frac{f}{\text{Hz}} \right)^{-4} \left| \sum_{m \neq 0} T_m(f) \right|^2 \right]^{-1/2} \quad (15)
 \end{aligned}$$

where:

$$T_m(f) = \frac{f^2}{f_m^2 - f^2 + i f_m f / Q_m} . \quad (16)$$

As noted in Section 3.2, we take normal mode frequencies f_m and quality factors Q_m based on [20] and calculated with the code [21], and apply a smooth cutoff factor

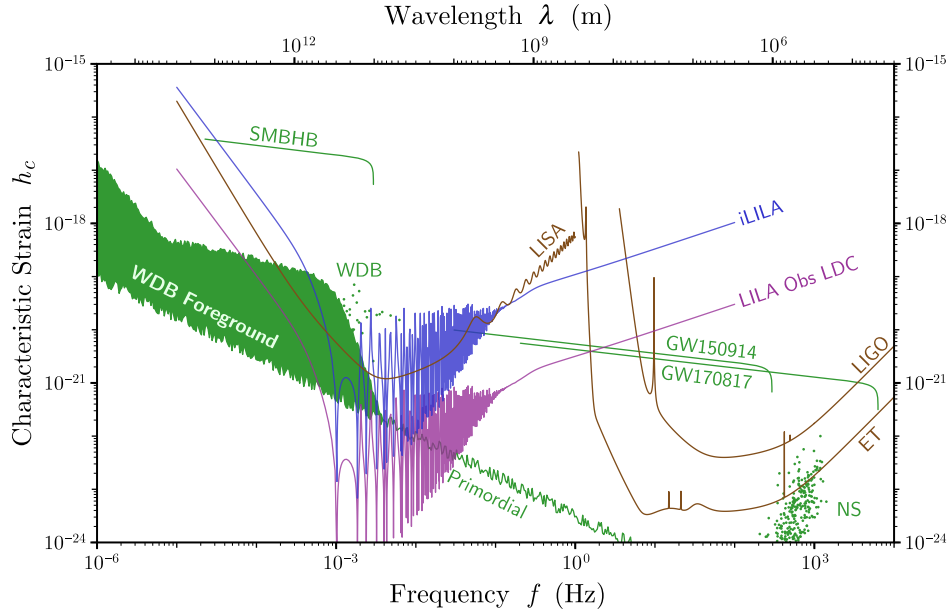


Figure 3. Characteristic strain sensitivity for several current and planned gravitational-wave detectors. The blue and purple curves show the expected sensitivity of the proposed iLILA (L-shaped 1 km) and LILA Observatory LDC (triangular 40 km) detectors described in the text. Lunar response is based on power-law fits to numerical models.

$e^{-f_m/0.05 \text{ Hz}}$ to Q_m to account for the expected reduction of mode coherence due to scattering in the megaregolith. The resulting sensitivity is plotted in Figure 3, under two different design assumptions:

- iLILA: The target sensitivity for the initial LILA mission, assuming a 1 km right-angled ($\theta = 90^\circ$) baseline, coating-dominated thermal noise $S_{x,T}(1 \text{ Hz}) = 1.2 \times 10^{-33} \text{ m}^2/\text{Hz}$, and high-power 500 mW photodiodes.
- LILA Observatory LDC: The low-frequency component of the LILA Observatory. As iLILA but with a 40 km triangular ($\theta = 60^\circ$) baseline.

Figure 3 also includes for comparison the design sensitivity of the current LIGO detector [22], the planned sensitivities for Einstein Telescope (ET) [23] and the Laser Interferometer Space Antenna (LISA) [24], and the characteristic signals from several representative sources:

- Known neutron stars (NS) [25], assuming non-axisymmetric ellipticity 10^{-9} and one year of integration.
- Inspirational events GW150914 (a black hole binary) and GW170817 (a neutron star binary), showing the last year of inspiral.
- A supermassive black hole binary (SMBHB) of two $3 \times 10^6 M_\odot$ black holes at 10 Gpc, again showing the last year of inspiral.
- Known white dwarf binaries (WDB) [26] assuming one year of integration, and an estimated confusion limit (WDB Foreground) from other such binaries in our Galaxy assuming 1/year frequency resolution.

- A standard Harrison-Zel'dovich (Primordial) gravitational-wave background from the Big Bang singularity. We note that verification of a stochastic signal will typically require two independent detectors at this sensitivity.

5. Conclusions

This paper has presented basic design parameters and sensitivity estimates for a Lunar gravitational-wave detector using unsuspended laser strainmeters. The proposed system is based on established technologies already developed for Terrestrial observatories, and can be deployed with modest astronaut participation at arbitrary locations on the Moon.

As shown in Figure 3, by harnessing the power of precision laser ranging along with resonant amplification by the Moon itself, we can achieve sensitivities competitive with orbital gravitational-wave detectors at millihertz frequencies, while also opening an important window into the decihertz band below the range of Terrestrial detectors. This will facilitate important science goals, from long-duration observation of stellar, intermediate, and supermassive binary systems, to probing the primordial singularity itself. The Moon promises to be an exciting new platform to revolutionize our understanding of the Universe.

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